

Total Eclipse on August 12, 2026: observations in Spain and prediction with COCONUT



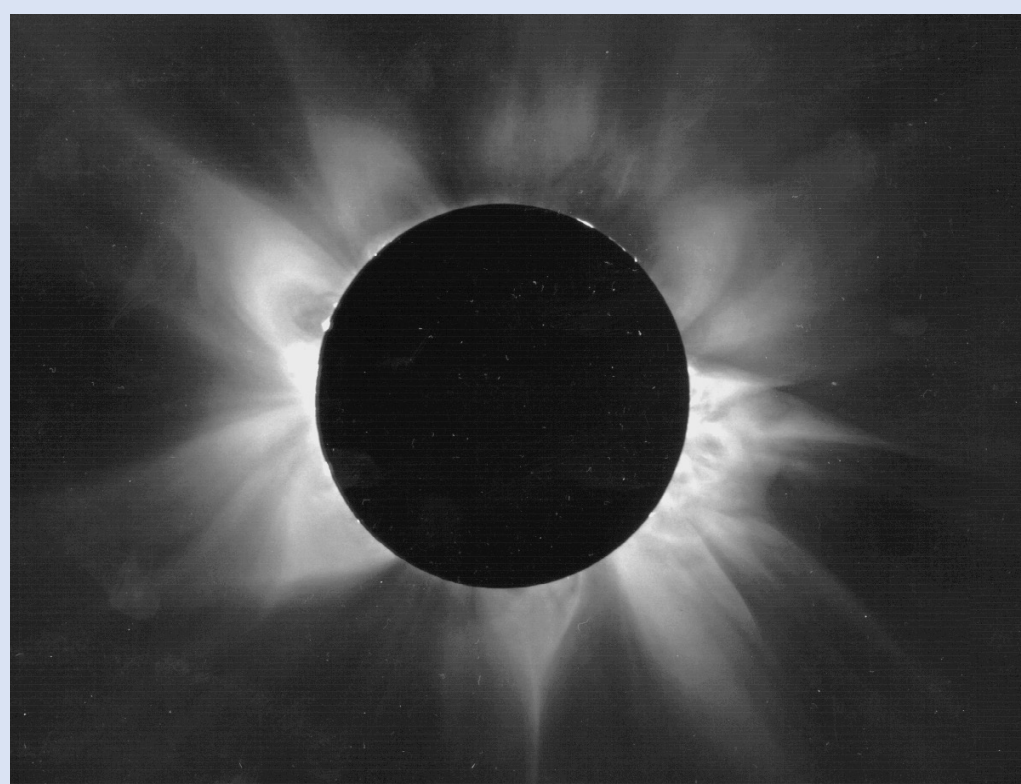
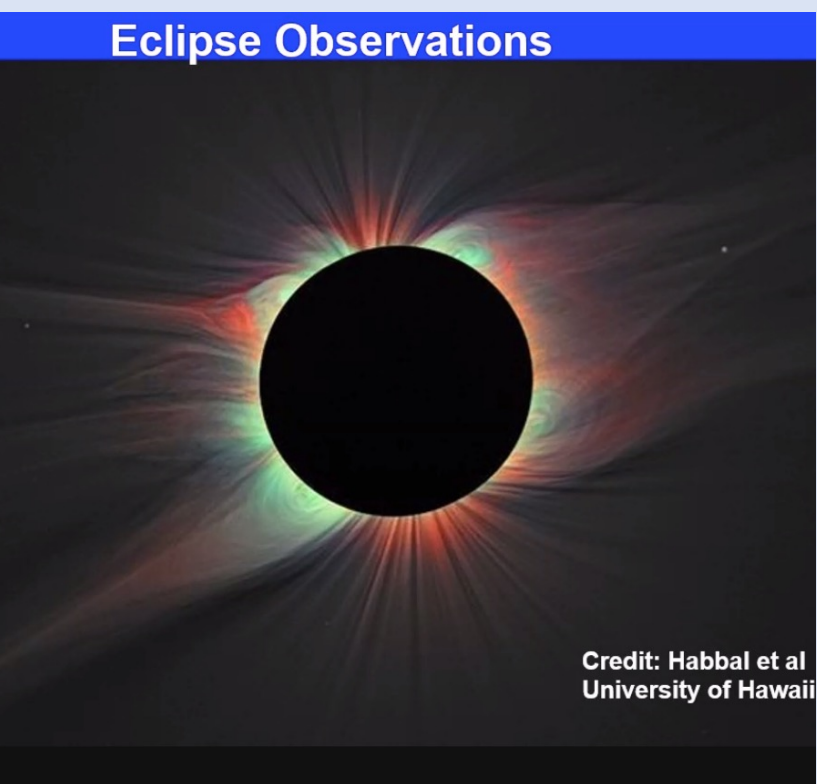
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Abstract: Total solar eclipses are a standard way of assessing coronal structures in the low solar corona (1-1.5 Solar Radius) and validating coronal models. Total solar eclipses, however, constrain the validation period of coronal modelling, as they occur rarely (roughly once per year somewhere on Earth). However, it used to be the only way to distinguish features in the low corona near the solar surface. Nowadays, the Proba-3 mission creates artificial total solar eclipses in orbit and thus provides observations of the low corona. Natural total solar eclipses provide a single snapshot of the solar corona, whereas Proba-3 can hold its artificial eclipse for up to 6 hours. This is very useful for validating time-dependent simulations which require continuous white-light observations. A program of space observations has been developed for the eclipse of 12 August 2026 using STEREO, Solar Orbiter, Aditya, Proba-3, as well as balloon experiments and flights to support the ground based multi-site observations in Spain obtaining the best coverage of the next total solar eclipse. COCONUT has been utilised to predict the previous total solar eclipse in April 2024 (Baratashvili et al. 2025, A&A). In the setup demonstrated in the manuscript, a low-resolution, simplified approach is used. However, multiple developments in the COCONUT model since the previous total solar eclipse allow continuous time-dependent and high-resolution simulations (Wang et al, 2025) for the predictions on the upcoming total solar eclipse on August 12, 2026, at 18:30 UT. Additionally, the high-quality images obtained before the eclipse by Proba-3 and the other space coronagraphs will be used for validating the predictions performed by the COCONUT model. Synthetic white-light images will be generated from the COCONUT simulations to compare directly with the observed images. This way we can use the total solar eclipse on August 12, 2026, to validate the COCONUT model, and identify its strengths and weaknesses.

Observations of the solar corona

The solar coronal temperature ranges from 0.8 to 10 million degrees. The solar corona is observed in Space by satellites such as SDO in X-rays or in white light with a coronagraph (SOHO/LASCO). On the ground, we can observe the solar corona in red/green lines at mountain observatories and in White light during eclipses, because the corona is too faint in the optical range (10^{-6}).



Green and red lines

Visible coronal lines

Bengt Edlén (1943)

Green coronal line (530.3 nm) from Fe¹³.

Green (530.3 nm) & Red (837.5 nm) coronal lines — Forbidden lines in the spectra of Fe XIV (Fe¹³) & Fe X (Fe⁹)

Fe X (Fe⁹) line

Fe XIV (Fe¹³) line

530.3 nm

837.5 nm

530.3 nm

837.5 nm

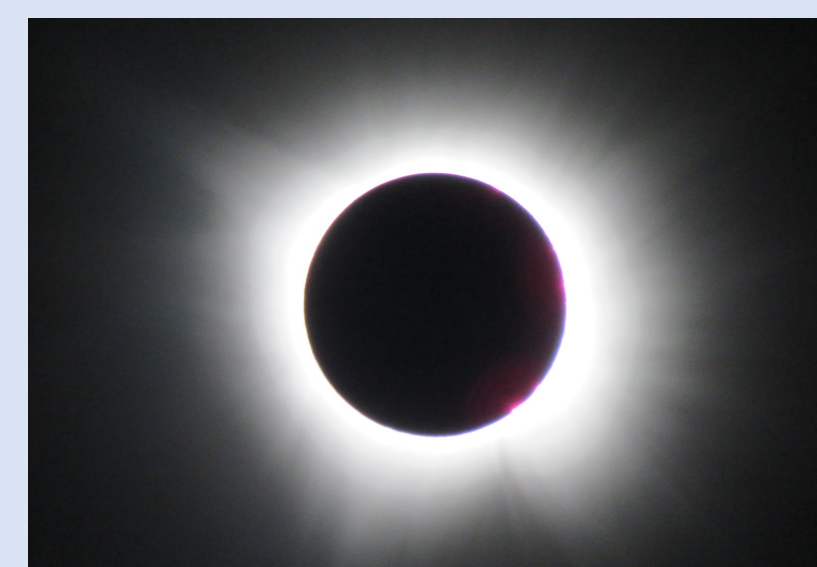
Fe X and Fe XIV coronal lines.

White light

Compton Diffusion: from the photosphere by the electrons of the corona (K) by the dust (corona F)

Morphology of the corona

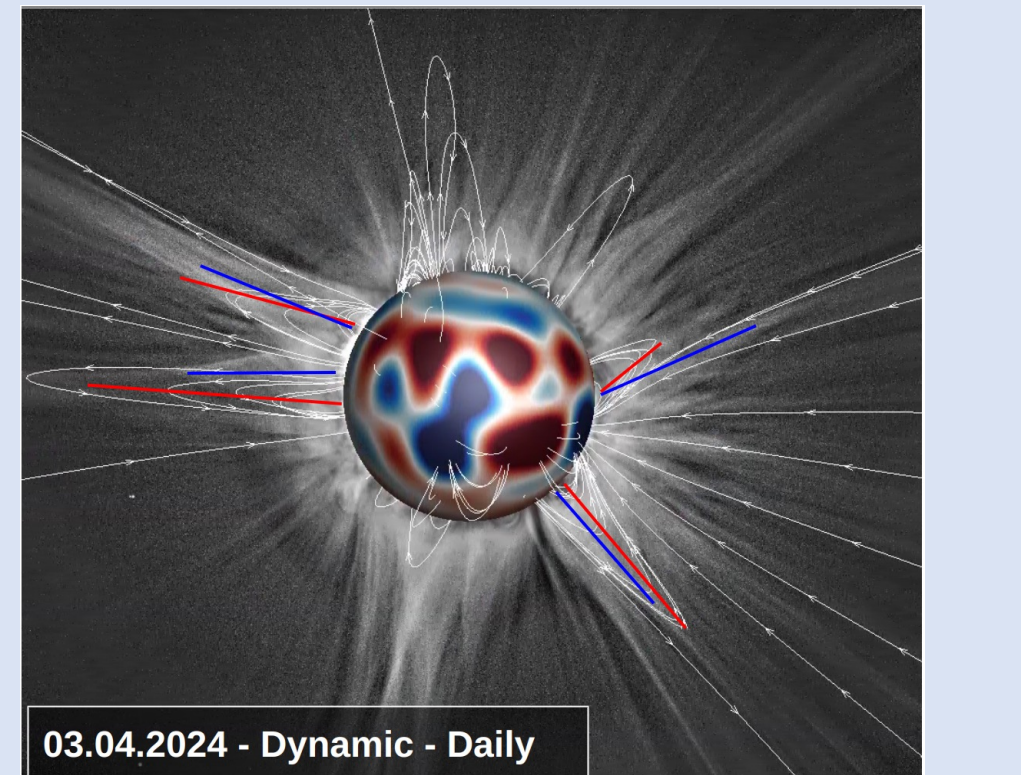
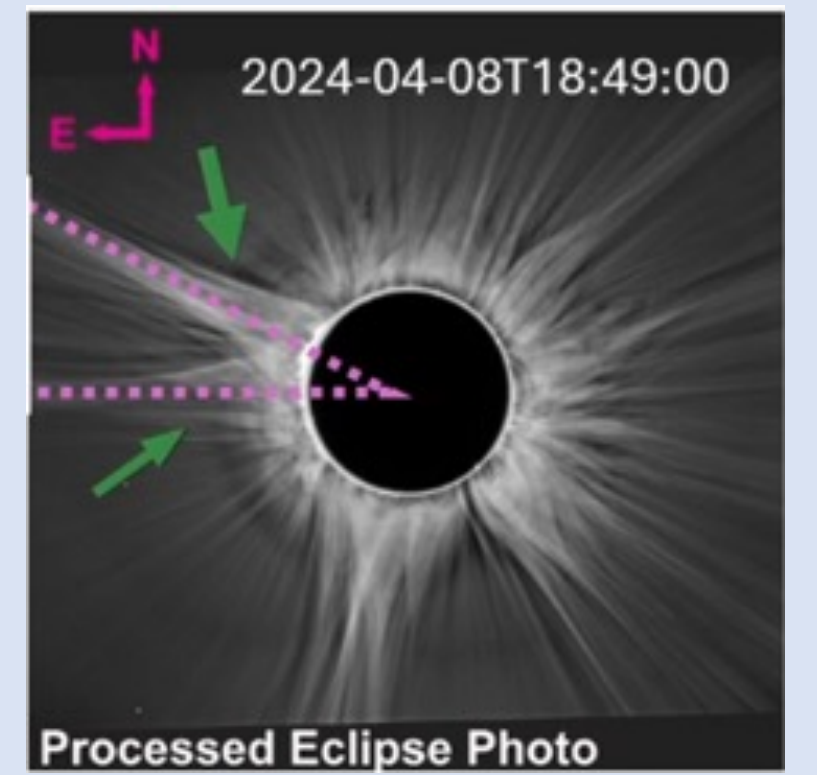
The morphology of the corona in eclipse depends strongly on the solar cycle phase: at solar maximum, the low corona is marked by the active regions which are mainly located between the latitudes 0 and 60 degrees). In the higher corona it is modulated by the Solar Wind, streamers and pseudo-streamers.



Eclipse of 8 April 2024

Left : Credit: top B.Schmieder, bottom: Eclipse team of Nanjing University - Wu, Sizhe (photographer); Li, Yihua; Huang, Yuhao; Lao, Qinghui; Cheng, Xin; Qu, Zhongquan.

Right: MHD simulations of KU Leuven (Baratashvili + 2026)



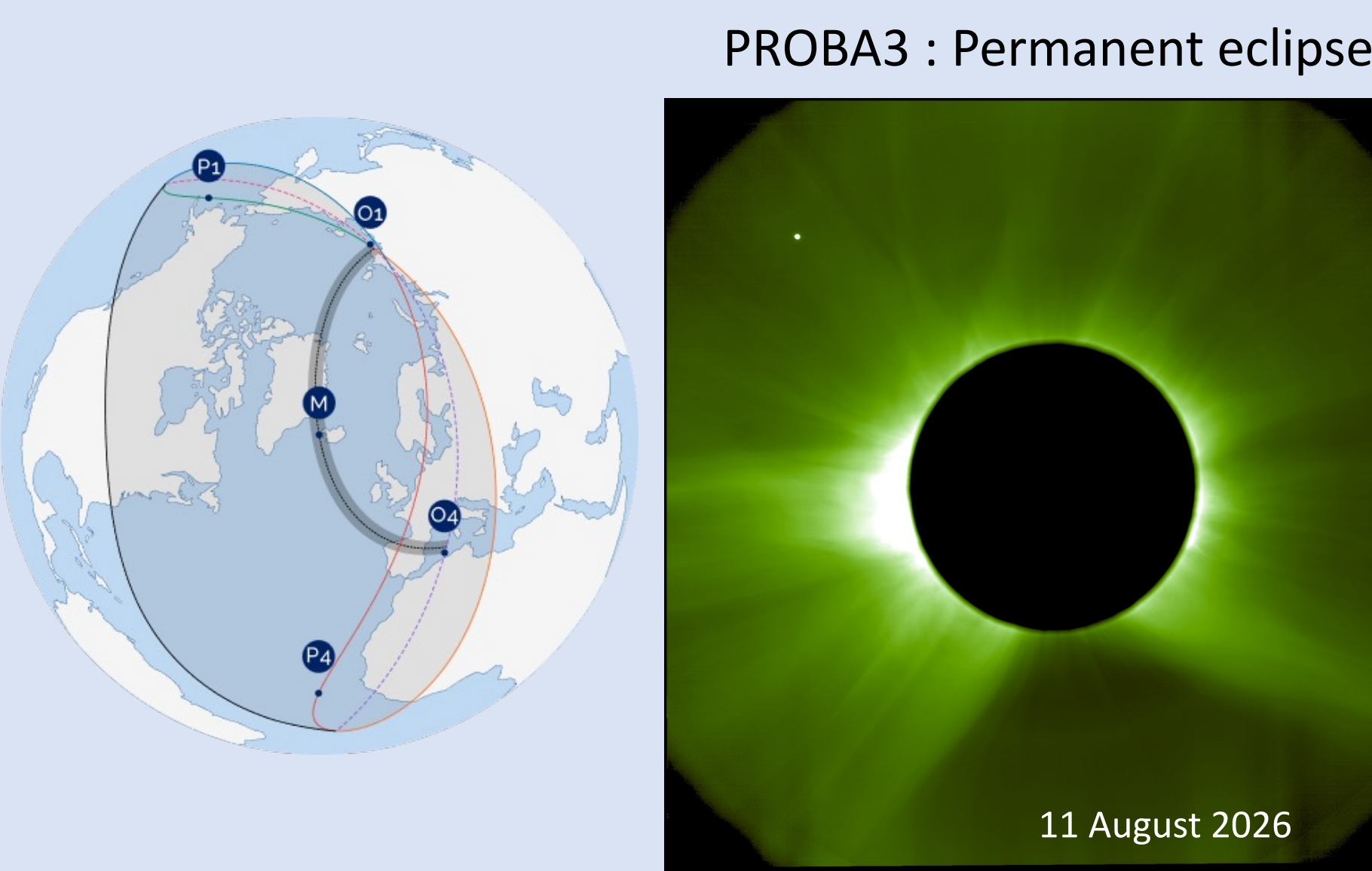
The image of the eclipse is overlaid in the background. The solar surface is colored with the processed radial magnetic field values. The directions of the streamers are compared from the structures in the image to the modelled streamlines in COCONUT input data, assimilation, and modelling of solar photospheric magnetic fields. (Baratashvili et al 2026)

Eclipse 12 August 2026



http://xjubier.free.fr/en/site_pages/solar_eclipses/TSE_2026_GoogleMapFull.htm?lat=42.33193&lng=-3.67130&elv=936.0&zoom=6&lc=1

The weather: <https://eclipsophile.com/tse2026/>



Preparation phase for Predictions for the eclipse

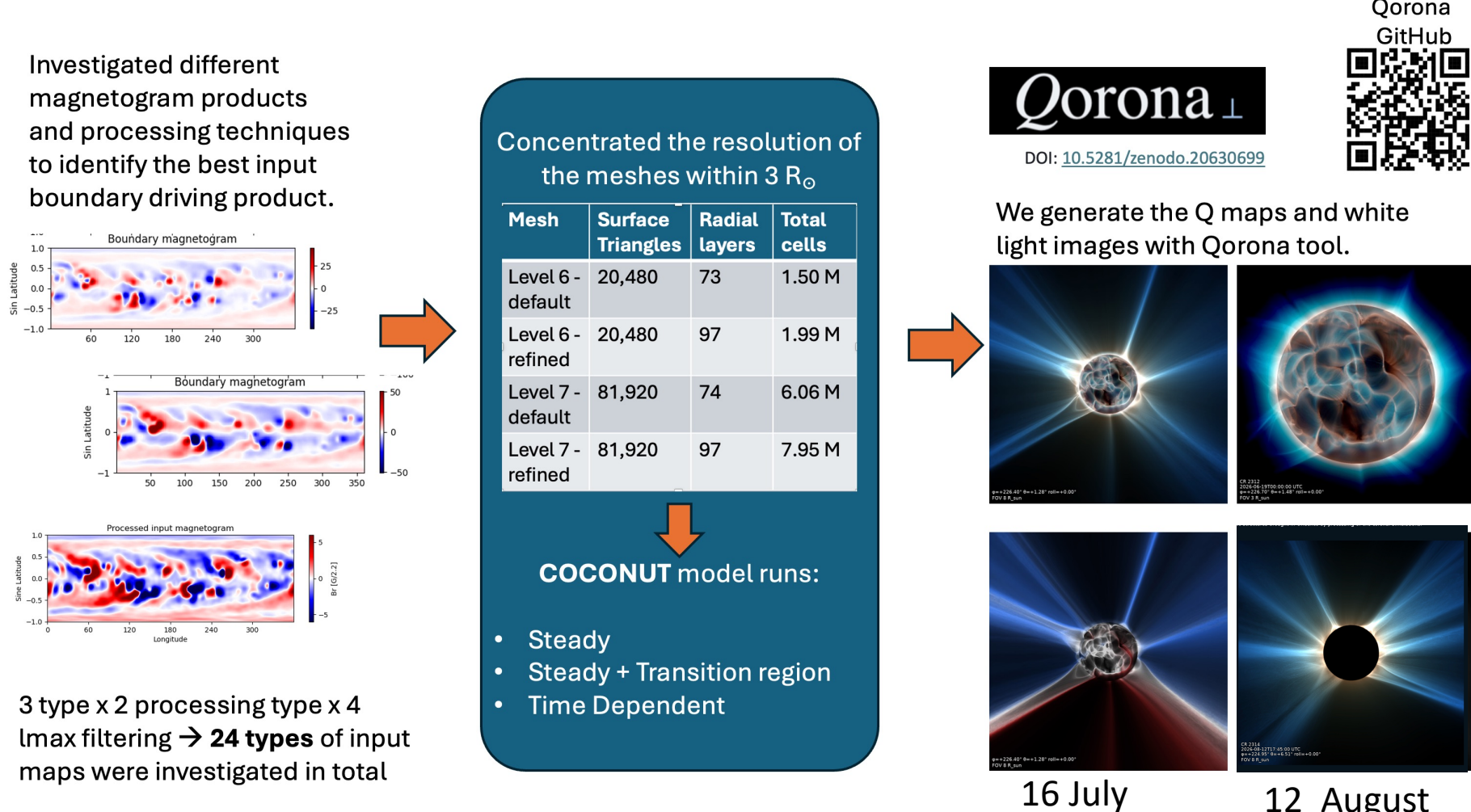


Figure 4 explains the different steps necessary to be done in the preparation phase. Synoptic magnetograms obtained from ground based observatory (GONG) or space data e.g. GONG, HMI and Solar Orbiter/PHI FDT respectively, are used and pre-processed. In parallel, AI-generated magnetograms are obtained using a database comprising far-side observations and surface flux transport model predictions (Jeong et al. 2026). Full MHD simulations of COCONUT with different adaptive meshes and heating terms are tested. This leads to the generation of level 6, 7 operational simulations with time dependent magnetograms or not. Forward modelling leads to white light synthesized images (Figure 4 right top panel). in COCONUT input data, assimilation, and modelling of solar photospheric magnetic fields (Dhib et al 2026).

The predictions with the 3D MHD coronal model **COCONUT** started on June 19. The predictions can be performed under both steady and dynamic regimes. GONG, GONG-ADAPT and HMI magnetograms were used for driving the coronal simulations. For this eclipse HMI-FDT-ADAPT was used. The enclosed boxes on the right denote areas that showed variations one solar rotation before the eclipse (16 July) and for the total solar eclipse.

Predictions with COCONUT @ KU Leuven

COCONUT (Perri&Leitner et al. 2022) is a 3D MHD coronal model accounting coronal heating, radiative losses and thermal conduction to accelerate solar wind (Baratashvili et al. 2025c). We are using this model to predict the total solar eclipse configuration on August 12, 2026. The daily predictions start July 16, one Carrington Rotation before the eclipse date.

Comparison between two solar rotations

Total Solar Eclipse on 12 August 2026: Preliminary Results

Original HMI map

HMI & AI map

Observations

Comparison of the COCONUT model results (squashing factor coloured by magnetic polarity) with coronal observations. (Courtesy of Hyun Jin Jeong)

Observations in Spain

Nanjing group at Hontoria de Cerrato (courtesy Bitao Wang)

A' Corona , at Playa Magdalena (courtesy Haopeng Wang)

Observations : Good predictions !

Courtesy B.Schmieder in Jadraque 2026-08-12 at 8:31 UT

Processed image at Belbimbre (Courtesy of K.Molnar)

Conclusion: The path is going from Iceland to Spain. The duration of the eclipse is 1:42 minutes. All the co-authors of the poster are part of a LUNEX-EGU-COSPAR-STP16 network of 50 observers and theorists. They were in contact and exchange eclipse data. The predictions are good.

References:

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